S&P Global & CRISIL Campus Hackathon 2026

AI/NLP Risk Engine: From Financial News to Portfolio Signals

End-to-End Tactical Index Rebalancing with Transparent Explainability

CANDIDATE NAME	CAMPUS / COLLEGE	EMAIL ID	DOWNSTREAM APPLICATION
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PROBLEM STATEMENT & SCOPE

Turning Unstructured Financial Noise into Machine Signals

Bridging financial NLP directly into systematic risk management and portfolio decisions

The Industry Challenge

- **Information Overload:** Thousands of headlines, SEC filings, and tweets are generated daily; humans cannot digest them in time.
- **Black-Box Dilemma:** Quantitative risk desks reject raw LLM/transformer outputs without explainable decomposition.
- **Execution Latency:** Portfolio adjustments must be systematically disciplined by deadbands, bounds, and turnover caps.

Core Approach in Four Pillars

- **Explainable by Design:** Impact score (1–10) is a weighted sum of 4 named components.
- **Measured, Not Assumed:** Evaluated on 4,838 PhraseBank headlines and 180 hand-labeled texts.
- **Empirically Validated:** Backtested against actual stock price returns using event studies.
- **100% Offline Reproducibility:** Bundled deterministic sample datasets require zero API keys.

Signal Specification Output

Every processed news article or tweet emits a structured **RiskSignal**:

1. Sentiment Score [-1.0, +1.0]
Domain-tuned FinBERT inference with confidence score

2. Event Classification (8 Classes)
Geopolitical, Macro, Credit, M&A, Product, Regulatory, Earnings, Other

3. Severity Impact Score [1 to 10]
Event severity (35%) + Sentiment strength (30%) + Source credibility (20%) + Volume (15%)

ARCHITECTURE & DATA FLOW

End-to-End System Architecture

From multi-source ingestion to FastAPI replay and interactive React dashboard



Pipeline Architecture

- **Ingestion:** Twitter Kaggle (2021–22), NewsAPI (2026), GDELT 2.0, yfinance daily prices.
- **Core NLP Engine:** Normalization, deduplication, cashtag entity linking, FinBERT, event classification.
- **Downstream Module A:** Daily EMA aggregation, conviction tilt, turnover cap, weight boundaries.
- **FastAPI & SSE Replay:** Asynchronous event-driven streaming with chronological ordering.
- **Vite + React UI:** 3-column workspace (News Stream, Signal Inspector, Portfolio Reaction).

NLP RISK PIPELINE DETAILS

Entity Linking, Sentiment & Explainable Impact Scoring

Strict entity resolution, domain FinBERT, and auditable linear impact scoring

Entity Linking & Attribution

- **Cashtag & Alias Matching:** Disambiguates symbols (e.g. $DIS, $AAPL) and company aliases.
- **Cross-Label Deduplication:** Resolves scraped duplicate tweets across tickers (preserving clean attribution).
- **Broadcast Tweet Handling:** Identifies multi-ticker tweets and scales impact proportionally.

FinBERT Sentiment Analysis

- Pretrained **ProsusAI/finbert** fine-tuned on financial phraseology.
- Generates signed score $s \in [-1.0, +1.0]$ with softmax confidence.
- Persistent disk cache for fast zero-overhead deterministic replay.

Explainable Impact Arithmetic

Configured in **config/engine.yaml** as a strict linear combination:

```
Impact = 1.0 + 9.0 * (
  0.35 * EventSeverity +
  0.30 * SentimentStrength +
  0.20 * SourceCredibility +
  0.15 * VolumeScore
)
```

- **Event Severity:** Regulatory/Credit = 0.9, Earnings = 0.7, Product = 0.5.
- **Source Credibility:** GDELT/News = 0.85, Verified NewsAPI = 0.80, Tweets = 0.50.
- **Auditability:** Every single score can be broken down in the UI's "Show the math" modal.

MODULE A REBALANCER

Tactical Index Rebalancer: Systematic Rules & Math

14-asset US mega-cap index dynamically tilted by sentiment momentum

Daily Scoring & Momentum Smoothing

- **Daily Aggregation:** Scores aggregated across ticker-day using impact-weighted sentiment.
- **Asymmetric Negative Weighting:** Negative news receives 1.25x weight multiplier (loss aversion).
- **EMA Smoothing:** 2-day half-life exponential decay prevents whipsaws:

  `EMA_t = alpha * Score_t + (1 - alpha) * EMA_{t-1}`
- **Deadband (+/-0.20):** Filters out low-conviction noise around neutrality.

Index Universe (14 Mega-Caps)

TSLA, AAPL, MSFT, AMZN, META, GOOGL, NFLX, AMD, PG, KO, DIS, BA, COST, PYPL (Equal Base Weight: 7.14%).

Allocation & Risk Controls

- **Conviction Tilt ($k = 1.5$):**

  `w_raw = w_base * (1.0 + k * EMA)`
- **Strict Boundary Box:** Weights clamped within $[0.5 \times w_{base}, 2.0 \times w_{base}]$:

  `3.57% <= Weight_i <= 14.29%`
- **Turnover Cap (10% daily):** Prevents transaction cost erosion from high-frequency reallocations.
- **Transaction Frictions:** Institutional 5 bps deducted on all rebalanced turnover.

Operational Safeguards

If market data is missing or signals drop to zero, portfolio gracefully reverts towards equal-weight benchmark.

EMPIRICAL VALIDATION

Abnormal Return Event Study: Testing Signal Efficacy

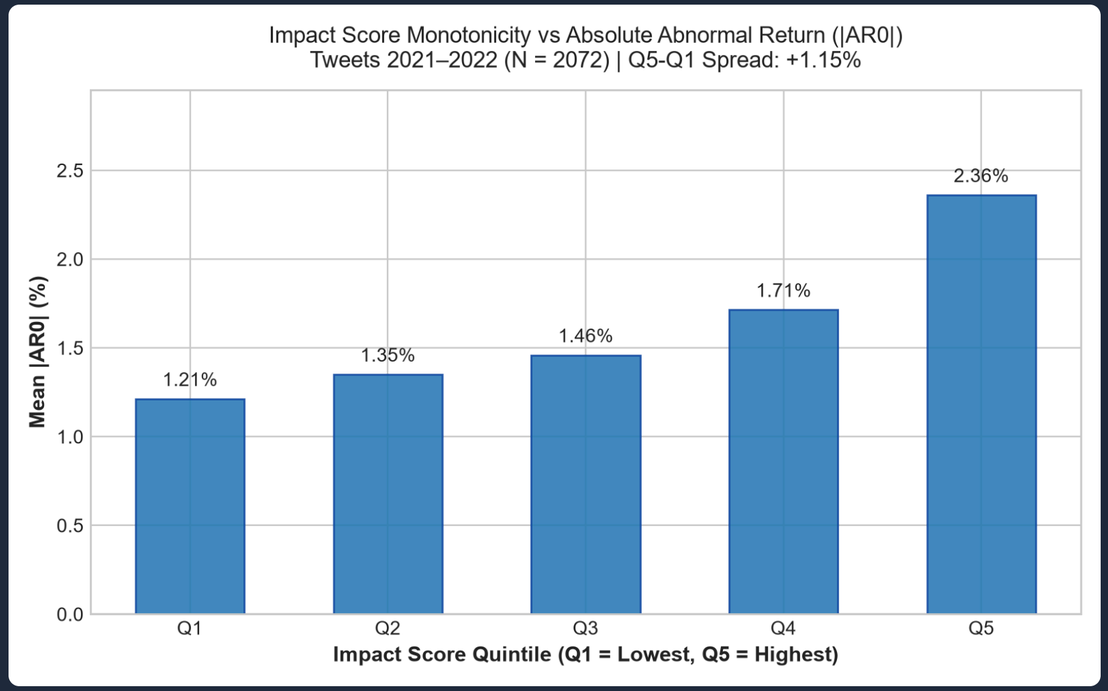
Rigorous statistical validation against 2,072 ticker-days of price returns

Key Statistical Findings

- **Spearman Rank Correlation:** $\rho = 0.159$ ($p = 3.08 \times 10^{-13}$, 95% CI $[0.122, 0.194]$) between impact score and abnormal price move $|AR_0|$.
- **Monotonic Quintile Scaling:** Absolute abnormal return scales monotonically from **1.21% (Q1) to 2.36% (Q5)**.
- **Top vs Bottom Spread:** $+1.15\%$ (95% CI $[+0.81\%, +1.57\%]$).
- **Robust to Confounders:** Partial correlation remains 0.116 controlling for volatility; survives TSLA exclusion ($\rho = 0.121$).
- **Placebo Permutation Test:** 1,000 within-ticker shuffles yield empirical $p = 0.001$.

Sentiment Directionality

Negative sentiment signals preceded a **-1.11%** 3-day abnormal return ($CAR(0,2)$), compared to $+0.08\%$ for positive signals (Long-Short spread $+1.19\%$).



BACKTEST & HONESTY REPORTING

Module A Backtest Results: Honest Reporting

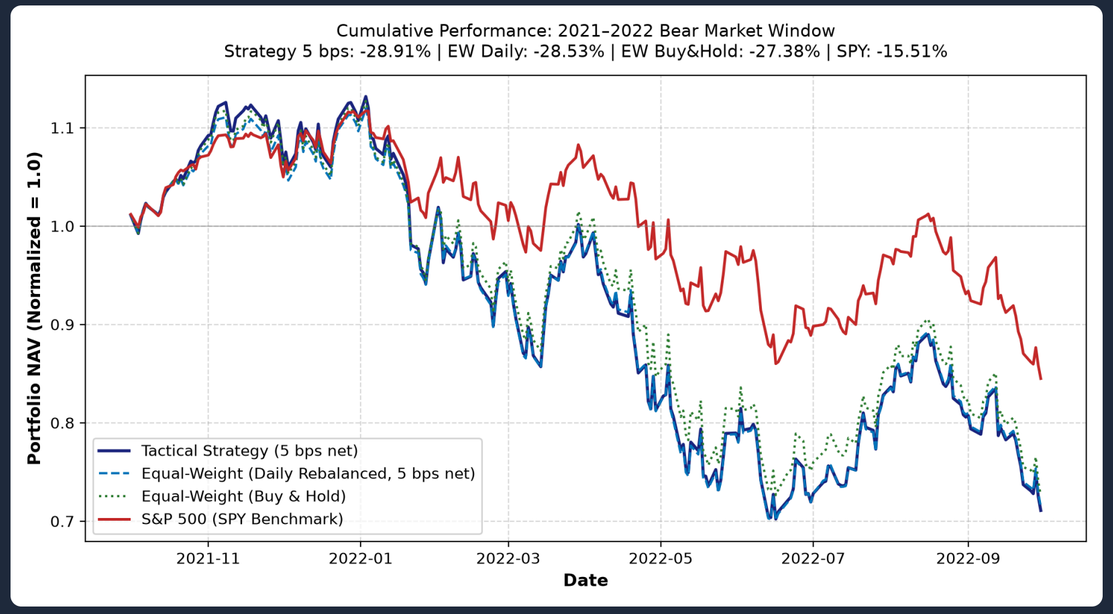
1-year simulation (Oct 2021 to Sep 2022) during the tech macro downturn

1-Year Performance Comparison

STRATEGY	CUM. RETURN	SHARPE	MAX DD
Tactical (5 bps Net)	-28.91%	-0.92	-37.95%
Equal Weight (Daily)	-28.53%	-0.91	-37.39%
Equal Weight (B&H)	-27.38%	-0.93	-35.56%
S&P 500 (SPY)	-15.51%	-0.71	-24.37%

Honest Diagnostic Takeaways

- **No Outperformance Claimed:** The strategy tracked equal-weight in a macro bear market (Placebo percentile: 48.5%).
- **Flat Sensitivity Grid:** Across tilt $k \in [0.75, 3.0]$ and half-life $[1, 5]$d, returns remained flat between -28.6% and -29.2%.
- **No P-Hacking:** Parameters pre-declared and frozen prior to evaluation.



EVALUATION & ROADMAP

Model Limitations, Learnings & Domain Impact

Transparent appraisal of current bounds and actionable future extensions

Identified Model Limitations

- **Benchmark vs Real World Gap:** FinBERT achieved 88.9% on FinancialPhraseBank, but dropped to 61.1% on real social chatter (tweets: 53.5%).
- **Positive Recall Deficit:** FinBERT missed subtle positive guidance (positive recall: 0.30 vs negative precision: 0.85).
- **Single Market Regime:** 2021–2022 was dominated by macro rate hikes, dampening single-stock alpha.
- **Class Imbalance:** Credit Event is naturally rare in mega-cap equity feeds (8 signals).

Enterprise Domain Impact & Module B

- **Immediate Desk Utility:** Delivers structured, auditable inputs for risk management and real-time news monitoring.
- **Module B Readiness:** The exact same signal stream directly feeds wholesale loan stress testing (credit migration, LGD shifts).
- **Production Architecture:** Fully containerized with Docker, offline sample dataset, zero lookahead bias, and 173 backend + 4 frontend tests.

Submission Summary

Complete codebase, documentation, and prebuilt dashboard submitted for the S&P Global and CRISIL Campus Hackathon 2026.